

WEIYI HUANG

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EDUCATION

Cornell University, College of Engineering, Bachelor of Science

Expected May 2027

Electrical and Computer Engineering, Aerospace Engineering Minor

Relevant Courses: Computer Architecture, Digital Logic and Computer Organization, Introduction to Circuits, Introduction to Microelectronics, Probability and Inference for Random Signals and Systems, Embedded Systems (FA25)

Thermodynamics, Statics and Mechanics of Solids, Fluid Mechanics (FA25), Introduction to Spaceflight Mechanics (FA25)

Phillips Exeter Academy, High School, Exeter, NH

Aug 2020-Jun 2023

ENGINEERING EXPERIENCE

Cornell University AutoDrone, *Electrical Subteam Lead*

Sept 2024-Present

- Designing and implementing power, RX/TX, FPV, and power architecture for LawnDart, a custom UAS for breaking the world speed record. Troubleshooted INAV firmware to ensure successful flight tests and consistent benchmarks.
- Developed custom motor test stand consisting of stress force sensor and laser RPM measurement using Arduino IDE to evaluate flight requirements, including electrical system integrity and motor performance during max motor power.
- Appointed as Electrical Subteam Lead to lead architecture of 6+ UAS drones, based on prior success in system testing.

Cornell University SciFi Lab, *Research Assistant*

Apr-Dec 2024

- Calibrated and optimized an automated acoustic sensing system with custom PCB processing module, speaker and microphone module, and Bluetooth module for detecting physical movements of user via audio signal scalograms.
- Integrated a unique Python neural network to identify and classify real-time exercise data from scalograms, enabling a novel, hands-free workout assistant.

Cornell Laboratory for Accelerator-based ScienceS and Education (CLASSE), *Electrical Engineer Intern* **Jun-Aug 2024**

- Routed 7500+ feet of sensor cabling to incorporate mechanical and electrical equipment for new HMF laboratory.
- Integrated, verified, and validated 5 large-scale, mission-critical electrical systems using real-time diagnostics and legacy interface adaptation, following proper procedure and enabling safe lab commissioning. Tasks included wire terminations within front-end equipment to revision of existing control network at back-end control systems.
- Devised and implemented numerous hardware and structural workarounds to address problems within existing electrical interfaces: integrated 2 redundant CHESS Vacuum Gate Valve Control panels and HMF Primary Gate Valve Control panel into 2 new control chassis designed to leverage existing sensor infrastructure and reduce new wiring.
- Upgraded multiple critical operation systems including CHESS thermal chassis, CHESS OK chassis, and radiation control monitors to reflect S5/S6/S7 experimental hutches as part of the installation of the new HMF laboratory.

Cornell University Biomedical Device (CUBMD), *Product Development and Media Subteam Member*

Feb-Dec 2024

- Implemented a wireless pump control mechanism using Arduino IDE and Bluetooth protocol to enable remote actuation and app integration. Refined design with custom PCB hardware, ATiny85 control module, HC-08 bluetooth module, and standardized motor controller for seamless product experience in condensed form factor.
- Streamlined crucial interactive elements on team website to broadcast team mission and attract new sponsors and student applicants. Concentrated on an enhancement of information and organization of hyperlinks on the homepage.
- Produced end-of-year promo video highlighting team accomplishments and final product design, reaching 5000+ students and faculty and increasing recruitment applications by over 25%.

LEADERSHIP EXPERIENCE

Project Level the Field (Project LTF), *Chapter Founder and President*

May 2024-Present

- Founded and manage Cornell Chapter of Project LTF, a 501(c)(3) nonprofit connecting students from underprivileged communities with college application resources from undergraduates at top universities.
- Interviewed and selected 7 executive board members. Recruited and screened 12 more mentors within initial weeks.
- Collaborate and communicate daily with executive board, national board, and other campus chapters to execute mentorship pipeline and manage chapter-wide objectives, including hybrid monthly events.

SPECIALIZED SKILLS

Programming: Python (GAN machine learning, diffusion model, PyTorch, Tensorflow), C/C++, Java, Object-Oriented Programming, Bash Scripting, Command Line, RTL/Verilog, SystemVerilog (hardware verification, PyMTL), TinyRV2

Tools: Solidworks, Onshape CAD, ANSYS FEA, KiCad, LTSpice, Intel Quartus Prime, Linux, Git/GitHub, LaTeX, Excel and Microsoft 365 Suite, digital system and circuit design, directed/random testing, low-level buses, logic-analyzer debugging, microcontrollers, Raspberry Pi, Arduino, rapid prototyping, logic and spectrum analyzers, oscilloscopes, soldering

Leadership Initiatives: Eagle Scout, Three-Year The President's Volunteer Service Award, Two-Year Rowing Team Captain